



Abhishek Rajak

AI Engineer & Deep Learning Specialist
B.Tech Electronics and Communication Engineering
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PROFESSIONAL SUMMARY

AI/ML Engineer & CV Researcher specializing in Computer Vision (Image Enhancement, Vision Transformers) and Generative/Agentic AI, backed by a strong foundation in Data Structures & Algorithms and competitive programming in C/C++. Proficient in orchestrating stateful agent workflows, implementing Transformer architectures from scratch, and deploying scalable vision pipelines using YOLO and FastAPI. Passionate about applying frontier LLMs and strong algorithmic fundamentals to complex, real-world vision and multi-modal engineering challenges.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (ECE)	National Institute of Technology, Surat	8.58/10.0	2024-2028
Senior Secondary	+ 2 High School Barbigha	91.8%	2021-2023

TECHNICAL SKILLS

- **Programming:** C/C++ (LeetCode – 288+ problems solved), Python
- **AI/ML Frameworks:** PyTorch, Scikit-learn, OpenCV, LangChain, LangGraph, Langsmith
- **Deep Learning:** Transformers, CNNs, GANs, LSTM, RNN, RAG, YOLO
- **Tools:** NumPy, Pandas, Matplotlib, Git/GitHub, FastAPI, Docker, Jupyter, Streamlit
- **Coursework:** DSA, OOPS, DBMS, SQL, Machine Learning, Agentic AI, Computer Vision

KEY PROJECTS

- **Raindrop Clarity (NTIRE 2026 Challenge 14th Ranked Globally)** 2026
Dual-focused day-night raindrop removal pipeline with Restormer and NAFNet Report | GitHub
 - Engineered a **2-stage adaptive pipeline** utilizing **Restormer** for global denoising and **NAFNet** for luminance-based expert routing, eliminating cross-domain interference.
 - Boosted model robustness via **8-Way Test-Time Augmentation (TTA)**, achieving a top-tier Final Score of **34.52 (27.57 PSNR, 0.8210 SSIM)**.
- **FDCMformer: Multimodal SAR-Optical Cloud Removal Transformer** 2026
Enhanced ECRformer architecture with cross-modal fusion for satellite cloud removal GitHub
 - Engineered a **Bi-Cross-Modal Attention (BCMA)** stem fusing SAR and optical modalities via cross-covariance attention (**CrossXCA**) and a learned gate, adding only **9K params** while staying fully drop-in compatible.
 - Designed **spectral-domain losses** (SAM, FFT, perceptual) and **gated skip connections**, resolving **fp16 NaN instabilities** via fp32-safe gradients for stable training on **SEN12MS-CR**.
- **LegalMind: AI Contract Risk Analyzer** Jul. 2025 – Oct. 2025
Document intelligence system leveraging Legal-BERT and DeBERTa-v3 architectures Live App | GitHub
 - Achieved **97.74% accuracy** in risk detection via an ensemble of **Legal-BERT** and **DeBERTa-v3**, coupled with a custom **LangChain RAG** pipeline and local **FAISS** embeddings.
 - Operationalized zero-latency inference using **FastAPI** and integrated **Supabase** for secure cloud storage.
- **AI Travel Agent: Intelligent Trip Planner** 2025
Full-stack generative AI system featuring Llama 3 and real-time geospatial tracking GitHub
 - Architected a trip planner using **Llama 3-70b** and **LangChain** to generate optimized, budget-aware itineraries.
 - Built a **FastAPI** backend powering a Live Survival Map with real-time tracking and automated PDF exports.
- **Autonomous Rover Cone Detection (YOLOv8)** 2025
Real-time navigation system for the International Rover Challenge (IRC) GitHub
 - Trained **YOLOv8** on a **60GB dataset** for robust cone classification and calculated visual center steering targets.
 - Validated autonomous navigation vision system on hardware during the **International Rover Challenge (IRC)**.

EXPERIENCE

- **Head** 2026 – Present
SillyCon AI Labs LinkedIn
 - Led the **ML/DL vertical** of SillyCon AI Labs, mentoring **25–30 second-year students** through structured, hands-on sessions on core Machine Learning and Deep Learning concepts.